ML Project Proposal

Early Identification of At-Risk Secondary School Students (Pass/Fail Prediction)

Machine Learning Project | **Task 1** | BSIS-4A

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1. Dataset

- **Source:** UCI Machine Learning Repository, Student Performance dataset [1].
- **Licence:** CC BY 4.0 (free to use if we cite it).
- **Size:** 649 students (Portuguese class file) from two secondary schools, school year 2005 to 2006.
- **Features:** 30 columns about the student (family, study habits, absences, past failures, social life, health), plus grades G1, G2 and G3 (final grade).
- **Privacy:** Names are already removed. Because sex and family columns are sensitive, we will check results for each group.

2. Problem Statement

Schools often find out a student is failing only after the final grade. By then it is too late to help. Can we use information known early to spot students who may fail, so teachers can support them sooner?

- **Type:** classification (the answer is pass or fail).
- **Target:** fail if the final grade G3 is below 10, pass if 10 or higher.
- **Note:** G1 and G2 are almost the same as G3, so our main model will not use them.

3. Metrics

Our main score is the F1 score for the fail class, with recall, balanced accuracy and ROC AUC as support. Only about 15% of students fail, so accuracy alone would be misleading. Missing a student who will fail is worse than a false alarm, so recall matters most. We will report the average and spread across folds.

4. Baseline and Plan

- **Baseline:** predict “pass” for every student. This gets about 85% accuracy but finds no failing students.
- **Models:** logistic regression, random forest, and k-nearest neighbours or SVM.
- **Testing:** 80/20 split, then 5-fold cross-validation on the training part. The test set is used only once, at the end.

Reference

[1] P. Cortez and A. Silva, “Using data mining to predict secondary school student performance,” in Proc. 5th FUBUTEC, Porto, Portugal, 2008, pp. 5–12.